Simulating Analyst Forecast Behavior using LLMs

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Motivation

LLMs simulate economic expectations

LLMs act as simulated economic agents (*homo silicus*); experiment and survey expectations replicated at the individual level

Limited evidence in financial forecasting

unlike the earlier *trained-ML* (non-LLM) literature on earnings prediction, LLM evidence is *directional*, not point forecasts

⇓

This paper simulates an LLM as the analyst: 121,369 matched analyst–LLM pairs.

A framework for human and AI forecasting (Hansen et al. 2025)

$$f^A = \mathbb{E}\Big[f(x_t,\, z_t) \,\Big|\, x_t,\, z_t\Big] \qquad f^L = \mathbb{E}\Big[f(x_t,\, z_t) \,\Big|\, x_t,\, h_t\Big]$$

x_t: data we provide

- fundamentals, macro, prices (mostly public)

h_t: past human forecasts

- h^{peer}: peer median
- h^{prior}: own prior forecast

z_t: private information

- what we cannot provide to the LLM

h_t **may carry part of** z_t: the analyst's private information flows into his own prior, the peers' into their median, so conditioning on h_t narrows the information gap.

Metrics

- **Forecast error**

$$FE = \frac{\text{actual EPS} - \text{forecast EPS}}{\text{stock price, prior fiscal year-end } (P_{T-1})} \qquad FE < 0: \text{ over-prediction (optimism)}$$

- **Accuracy**: $|FE|$
- **Relative accuracy**

$$SUP = |FE^L| - |FE^A| \qquad SUP > 0: \text{ the analyst wins}$$

Data: Korean individual analyst forecasts (FnGuide, 2018–2025)

	N	Mean	Std	P25	Median	P75
Panel A: Forecast metrics						
Actual EPS	121,369	6,571	13,353	1,005	3,285	8,741
Analyst EPS	121,369	7,958	11,871	1,591	4,053	9,614
LLM EPS	121,369	8,044	11,581	1,650	4,150	9,700
Price P	121,369	122,107	197,836	28,300	59,300	142,100
$\lvert FE\rvert^{\text{Analyst}}$	121,369	0.0452	0.0867	0.0071	0.0198	0.0497
$\lvert FE\rvert^{\text{LLM}}$	121,369	0.0480	0.0879	0.0076	0.0211	0.0535
SUP	121,369	0.0029	0.0407	−0.0040	0.0006	0.0072
Panel B: Firm and forecast characteristics						
Size	113,839	15.10	1.51	14.04	14.96	16.10
BtoM	113,816	1.20	1.21	0.43	0.83	1.58
Leverage	114,116	0.48	0.19	0.32	0.49	0.62
Loss	111,752	0.15	0.36	0.00	0.00	0.00
DU	118,580	0.024	0.031	0.010	0.018	0.031

Placing the LLM at date *t*

We are not forecasting the future from a common date: the study **replicates forecasts issued 2018–2025**, so the LLM must stand at each historical forecast date *t*.

1. **A dated persona**: an analyst at {brokerage}, forecasting on {date}
2. **Point-in-time inputs**: every data block truncated at *t*
3. **Instruction**: the developer message forbids referencing anything after *t*
4. **No external search**: the API call carries no browsing or tools

This cannot be verified *deductively*: the weights may still encode later outcomes. §5.4 tests it *indirectly*, at two training cutoffs.

The prompt: an actual example (Samsung Electronics, 2024-02-01)

```
DEVELOPER MESSAGE
You are an equity analyst at Kyobo Securities, producing a
forecast on 2024-02-01.
Your most covered companies: 006400 (SAMSUNG SDI);
003670 (POSCO FUTURE M); 066570 (LG ELECTRONICS)
Your most covered sectors: Secondary battery mfg. (28202); ...
Your objective is to predict the realized EPS as
accurately as possible.
Do not reference or search for any events or
developments that occurred after this date.
Use only the information provided below.

USER MESSAGE
[0] COMPANY OVERVIEW
Company: SAMSUNG ELECTRONICS CO,.LTD (005930)
Sector: Communication and broadcasting equipment
manufacturing (KSIC 264)   Market: KOSPI
Market Cap (2024-01-31): 434,003,191 (KRW Millions)

[1a] DOMESTIC MACROECONOMIC DATA
           2020Q4  2021Q1  ...  2023Q3  2023Q4
GDP_YoY(%)    2.6     7.0  ...     1.5     2.2
CPI_YoY(%)    1.4     2.5  ...     3.1     3.4
Unemp(%)      5.0     3.9  ...     2.3     2.6
--- BOK Forecast (published 2023-11-30) ---
2024:  GDP 2.1%  CPI 2.6%  Unemp 2.9%  CA 490.0

[1b] MARKET DATA
           2020Q4  2021Q1  ...  2023Q3  2023Q4
Dubai                60.2  ...    86.6    83.7
KRW/USD            1114.8  ...  1311.7  1320.7
Spread_AA             1.1  ...     0.8     0.8
As of 2024-01-31: Dubai 82.41; KRW/USD 1334.6; ...

[2] FINANCIAL STATEMENTS (KRW Millions)
                FY2020       FY2021       FY2022
Revenue    236,806,988  279,604,799  302,231,360
Op. income  35,993,876   51,633,856   43,376,630
Profit(loss) 26,407,832  39,907,450   55,654,077
... (assets, equity, liabilities: 13 items)

[3] STOCK PRICE (Monthly Close, KRW)
2021-02  82,500  ...  2023-12  78,500  2024-01  72,700

[4] EPS DATA (as of 2024-01-31)
       Avg_Shares     Net_Income(1000)  Actual EPS
FY2020 6,792,669,250  26,090,846,000        3,841
...
FY2022 6,792,669,250  54,730,018,000        8,057

[5] MEDIAN OF OTHER ANALYSTS' FORECASTS TO DATE
       Median(KRW)  # Analysts
FY2023       1,613          27
FY2024       4,174          20

[6] YOUR PAST FORECASTS FOR THIS COMPANY
2023-05-09  FY2023  1,415
2023-07-28  FY2023    963
2023-09-26  FY2023  1,060
Predict the FY2024 annual EPS for this company.
Give the EPS in KRW first, then your confidence in
that EPS as an integer from 0 (none) to 100 (certain).
```

Simulating an analyst with an LLM

The call

- reasoning effort: `medium` (default)
- verbosity: `medium` (default)
- temperature/top-p: not settable on reasoning models

Two arms

- **Full**: the entire prompt
- **Ablated**: h^{peer}, h^{prior} removed

Two models, identical prompts

- **gpt-5.4-nano**: main results
- **gpt-5-mini**: robustness; every pattern repeats

Guarding against leakage

- inputs truncated at the forecast date
- the prompt forbids it: "Do not reference or search for any events … after this date"
- ex-post validation: §5.4

Results: overview

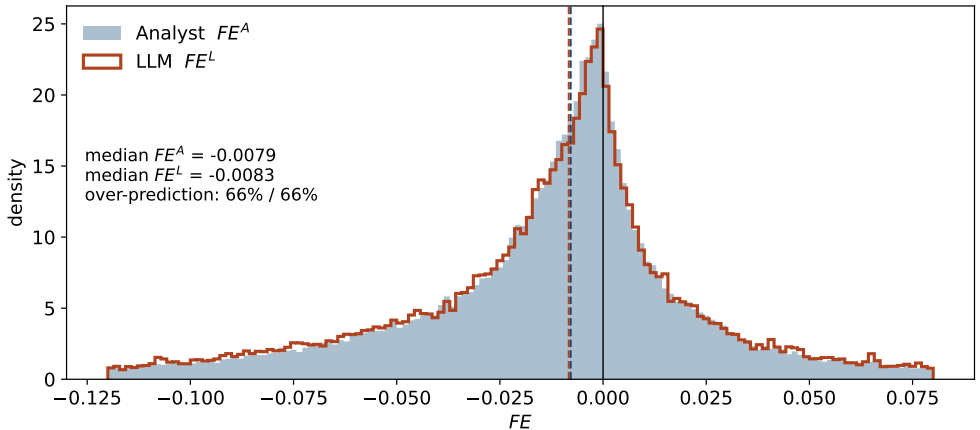
5.1
Forecast alike?

5.2
More accurate?

5.3
Confidence

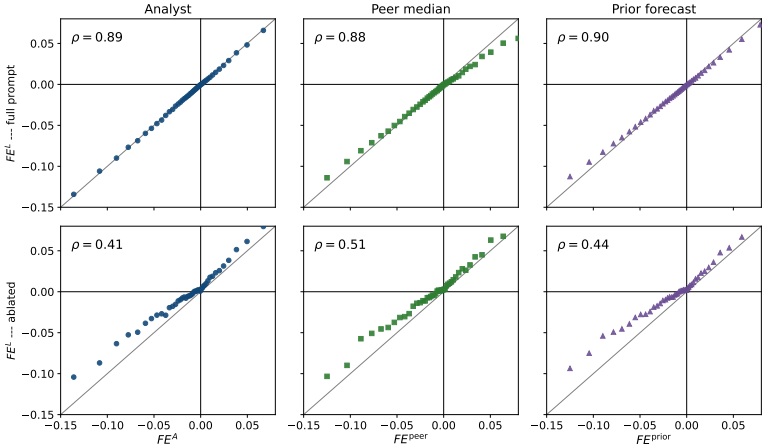
5.4
Recalling?

5.1 · Do they forecast alike?



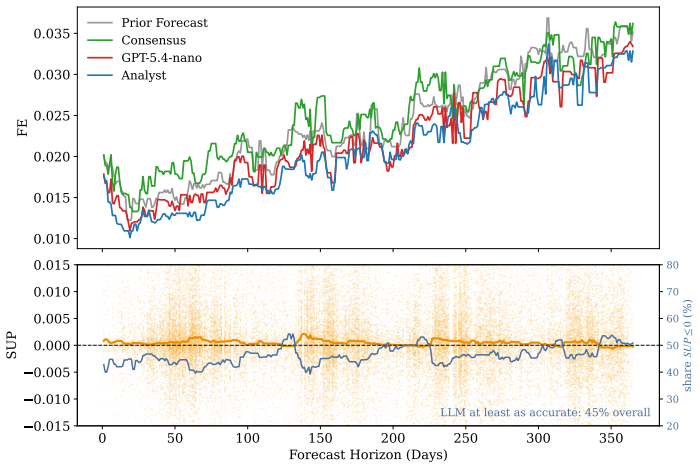
Both over-predict: the two error distributions nearly coincide, and the LLM inherits the analyst's optimism (analyst optimism: Easterwood & Nutt 1999).

5.1 · Error co-movement: the LLM vs. the analyst and h_t



The LLM strongly anchors on the human benchmarks: ablate them, and its similarity to the analyst's forecasts falls.

5.2 · Accuracy over the forecast cycle



The analyst is more accurate at every horizon; the LLM at least as accurate in 45% of forecasts.

5.2 · The gap, across the whole distribution

$\lvert FE \rvert$	P25	Median	P75	Mean
Analyst	**0.0071**	**0.0197**	**0.0498**	**0.0453**
LLM	0.0076	0.0211	0.0537	0.0482
gap	+7%	+7%	+8%	+7%
Prior forecast (input)	0.0084	0.0231	0.0584	0.0520
Peer median (input)	0.0089	0.0247	0.0634	0.0565
LLM (ablated: peer median & prior removed)	0.0119	0.0324	0.0803	0.0728

analyst < **LLM** < **inputs** < **ablated** across the whole distribution: the LLM **leans heavily on the human benchmarks** (peer, prior), yet **uses them to do better**, beating both; still 7–8% behind the analyst.

5.2 · Where does the gap sit?

$$y_{ijt} = \beta' X_{ijt} + \alpha_{\text{yr}} + \alpha_{\text{firm}} + \alpha_{\text{analyst}} + \varepsilon_{ijt}$$

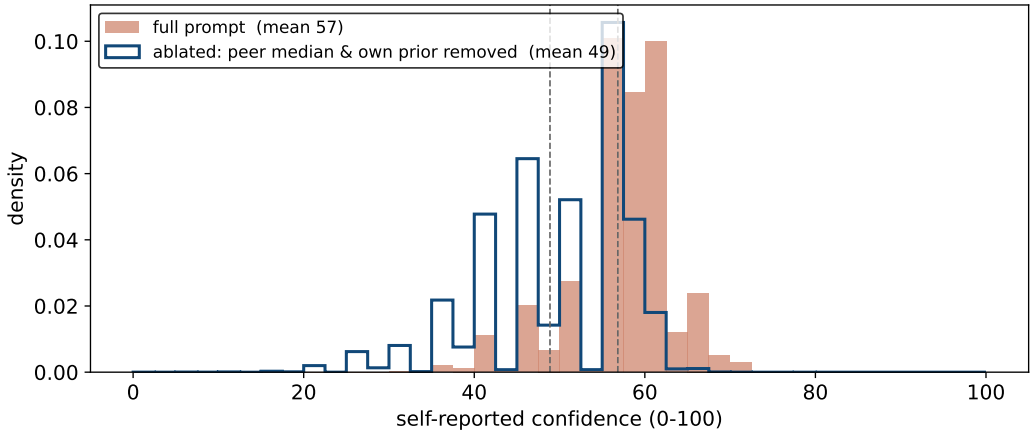
$y = \mathbf{1}\{SUP > 0\}$ (analyst wins) or SUP

$y =$	(1) $\mathbf{1}\{SUP > 0\}$	(2) SUP
Size	$+0.0316^{**}$ (2.01)	-0.0006 (-0.82)
BtoM	$+0.0074$ (0.75)	$+0.0001$ (0.11)
Leverage	$+0.0136$ (0.22)	$+0.0040$ (1.31)
Loss	$+0.0306^{***}$ (2.84)	$+0.0031^{***}$ (4.93)
DU	$+0.2877$ (1.23)	$+0.0379^{**}$ (2.56)
Horizon	-0.0213^{***} (-6.13)	-0.0008^{***} (-5.34)
ln(N)	-0.0210 (-1.12)	-0.0013 (-1.50)
Adj. R^2	0.028	0.050

The edge is largest for **loss firms** and **larger firms**, under **disagreement**, at **short horizons**.

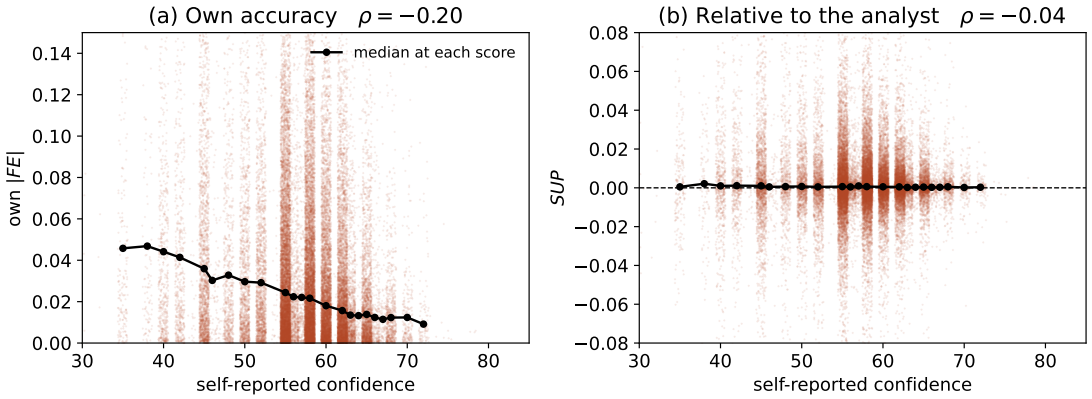
Median-based summaries throughout: analysts face absolute-error loss (Basu & Markov 2004; Gu & Wu 2003); firm-clustered t-stats, firm-block bootstraps.

5.3 · The confidence score



Remove the human signals from the prompt and confidence falls, 57 → 49: the self-assessment also reflects the prompt.

5.3 · Confidence vs. $|FE|$ and SUP



Confidence tracks how hard the forecast is, not where it stands.

5.3 · When is it confident?

$$z(\text{confidence})_{ijt} = \beta' X_{ijt} + \alpha \text{ (fixed effects)} + \varepsilon_{ijt}$$

	(1)	(2)
Size	$+0.054^{***}$ (3.1)	$+0.173^{***}$ (4.1)
BtoM	-0.053^{***} (-3.6)	-0.076^{***} (-2.9)
Leverage	$+0.020$ (0.2)	-0.562^{***} (-3.0)
Loss	-0.777^{***} (-20.1)	-0.480^{***} (-16.3)
DU	$+0.356$ (0.7)	-0.015 (-0.1)
Horizon	-0.155^{***} (-20.5)	-0.160^{***} (-21.3)
ln(N)	$+0.194^{***}$ (4.8)	$+0.336^{***}$ (6.6)
Adj. R^2	0.156	0.254
Fixed effects	Yr	Yr, Firm, Analyst

Confident for **profitable**, large, well-covered firms, **near year-end**.

5.4 · Forecasting or recalling?

Training cutoffs differ: **gpt-5-mini May 2024** · **gpt-5.4-nano Aug 2025** ⇒ FY2024 actuals are post-cutoff for mini; **FY2025 for both**.

	gpt-5.4-nano (main)		gpt-5-mini	
Target year	Median $\|FE\|$	Win rate	Median $\|FE\|$	Win rate
FY $\leq$ 2023	0.0204	45.1%	0.0206	43.1%
FY 2024 (clean: mini)	0.0239	45.5%	0.0243	43.6%
FY 2025 (clean: both)	**0.0224**	**46.6%**	0.0227	44.9%

Accuracy is slightly better after the training cutoffs. If the LLM were simply copying from its training data, this would not be the case.

Conclusion

Question	Answer
Do they forecast alike?	Yes: same optimism, errors 0.88 correlated; similarly close to the other human benchmarks
Which is more accurate?	The analyst, overall; the gap *narrows* for profitable firms, low disagreement, long horizons
Does it know its own confidence?	It tracks *how hard* the forecast is (-0.20), not where it stands against the analyst (-0.04)
Forecasting or recalling?	Forecasting: no break at either training cutoff

Further research

1. **Can larger models overcome the gap?**
   we test two small models: scale, reasoning effort, and fine-tuning remain open
2. **True out-of-sample**
   real-time forecasts of future fiscal years, beyond *every* training cutoff
3. **A Shapley decomposition of the prompt**
   remove each input block one at a time: which signals does the LLM weigh most when it forecasts?
4. **Other markets**
   replicate the design in other countries: U.S. individual analyst forecasts via I/B/E/S
5. **Conditionally combining the two forecasts**
   weight analyst and LLM state by state: can the pair beat the analyst alone?

Thank you

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